

Arena/Streams/Layers

metrics_episode() Contract

Pipeline Stages

DIVERGENT PATH (without contract):

mean_biomass

q75_biomass

**SINGLE
Interface**

```
def metrics_episode(  
  obs, actions, rewards  
) → metrics_dict
```

Optimizer

Tournament

Inference

HEAS UNIFIED PATH (with contract):

metrics_episode()

NO DIVERGENCE

Example: Policy A (Rank inconsistency)

Divergent:

Stage 1: mean_biomass = 50.5
↓ Rank #3

Stage 2: q75_biomass = 57.5 vs.
↓ Rank #7

HEAS:

All stages: mean_biomass = 50.5
↓ Rank #3 everywhere

Figure X: HEAS metrics_episode() Contract Architecture